NLP MINI PROJECT REPORT

NLP-Based Classification of Government Health Policy Press Releases

Classical NLP preprocessing and transformer-based text classification

Submitted by

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Abstract

This project develops an NLP workflow for categorizing official Government of India health-policy press releases into three topic labels: Launches, Platforms & Partnerships; Events, Hackathons & Conferences; and Progress Updates & Reviews. The workflow combines text normalization and interpretable linguistic analysis with two supervised document classifiers: a TF-IDF plus Logistic Regression baseline and a fine-tuned DistilBERT sequence classifier. NLTK is used for sentence segmentation, tokenization, POS tagging, POS-aware lemmatization, and WordNet semantic lookup; supplementary experiments also demonstrate morphology, shallow/deep parsing, word embeddings, contextual embeddings, named entity recognition, and zero-shot classification.

The initial labelled corpus contained 11 records. A further 23 official-source press releases were added, producing a 34-document working dataset. Its class distribution is 11 Launches/Platforms/Partnerships, 13 Events/Hackathons/Conferences, and 10 Progress Updates/Reviews. Five unmapped records from an earlier 16-record input were held aside instead of being silently assigned uncertain labels. A stratified split created 21 training, 6 validation, and 7 test documents.

On the seven-document test set from the 34-document dataset, TF-IDF plus Logistic Regression achieved 71.43% accuracy and 70.24% weighted F1. Fine-tuned DistilBERT achieved 42.86% accuracy and 35.92% weighted F1, and it did not correctly identify the Progress Updates & Reviews class in this test run. These results are useful for demonstrating an end-to-end workflow but remain preliminary because the dataset is small and starter labels require further review. The baseline's stronger result also shows that a pretrained transformer does not automatically outperform simpler methods when only a few dozen labelled documents are available.

Keywords: Natural Language Processing, health-policy press releases, text classification, NLTK, WordNet, TF-IDF, Logistic Regression, DistilBERT, Hugging Face Transformers, zero-shot classification.

Project at a Glance

Item	Current project configuration
Input	Full text of an official health-policy press release
Output	One of three release-topic labels
Working corpus	34 labelled records in the current run
Class counts	11 Launches/Platforms/Partnerships; 13 Events/Hackathons/Conferences; 10 Progress Updates/Reviews
Models	TF-IDF + Logistic Regression; fine-tuned DistilBERT
Primary evaluation	Accuracy, weighted precision/recall/F1, per-class report, confusion matrix, error review
Current strongest run	TF-IDF + Logistic Regression: 71.43% accuracy, 70.24% weighted F1 on 7 test documents

1. Introduction

Government departments publish press releases that announce new digital platforms, public programmes, partnerships, public events, implementation milestones, and progress updates. A reader who wants to monitor one specific type of update may need to scan lengthy documents manually. A text-classification system can organize these announcements into topic groups and make a release collection easier to search and browse.

Natural Language Processing (NLP) provides methods for turning unstructured language into more consistent and analysable representations. Sentence boundary identification and tokenization create units for processing; POS tagging and morphological analysis expose grammatical information; lemmatization normalizes inflected forms; WordNet provides lexical-semantic information; and TF-IDF or transformer embeddings convert text into numerical representations for machine learning.

1.1 Motivation

- Reduce the manual effort required to organize long government health-policy announcements.
- Compare a conventional sparse-text baseline with a pretrained transformer fine-tuned for the same classification task.
- Demonstrate how classical preprocessing and linguistic inspection complement supervised classification.
- Create a reproducible, inspectable workflow that can be extended as more labelled press releases become available.

1.2 Scope

The core project is single-label, three-class press-release classification. It does not determine whether a policy is effective, factually correct, or important; it only predicts which topic category best describes the release. Other laboratory experiments (such as morphology, HMM POS tagging, NER, contextual embeddings, and zero-shot classification) provide supporting demonstrations of NLP methods. They are not all concatenated into the main classifier input.

2. Problem Definition and Objectives

Problem statement: Given the full text of an official health-policy press release, predict which of three predefined release-topic categories best describes the document. The input is a raw text string from a JSON record. The output is one category label.

2.1 Input and Output

Element	Description
Input	A press-release record containing a source identifier/filename and the full announcement text.
Output	One label: Launches, Platforms & Partnerships; Events, Hackathons & Conferences; or Progress Updates & Reviews.
Task type	Single-label, three-class supervised text classification.
Primary models	TF-IDF + Logistic Regression baseline, compared with fine-tuned DistilBERT.

2.2 Illustrative Example

Input: “The Ministry of Health and Family Welfare launched a new digital health platform to enable citizens to access healthcare services online.”

Expected output: Launches, Platforms & Partnerships. This is a representative example of the intended label because the main subject is the introduction of a digital platform. A live demo using a similar health-platform paragraph was correctly classified in the notebook; this is a demonstration, not part of the held-out test metrics.

2.3 Objectives

- Load, validate, and maintain a JSON collection of government health-policy releases using official-source records.
- Normalize text and demonstrate sentence segmentation, tokenization, stop-word handling, POS tagging, lemmatization, and WordNet semantics.
- Train a TF-IDF + Logistic Regression baseline using the training split only.
- Fine-tune DistilBERT for three-class sequence classification and evaluate on an untouched test split.
- Compare model scores, inspect class-level errors, and document limitations caused by the small corpus.
- Provide a live prediction path for a newly entered press-release paragraph.

3. Dataset and Topic Labels

The dataset is stored as JSON records containing press-release text, a filename or source identifier, metadata, and an explicit label. The original input contained 16 records; the original filename-based labelling code mapped 11 and left 5 without verified mappings. The five unmapped records were quarantined for review rather than assigned labels by guesswork.

The dataset preparation package added 23 official-source press releases from the Press Information Bureau (PIB), with relevant cross-references from the official Ayushman Bharat Digital Mission (ABDM) press-release and Scan & Share pages. The completed run contained 34 labelled records: the 11 mapped original releases plus 23 additions. Each added record has a starter label based on the main subject of the release. Those labels should still be reviewed by a human, especially when a release combines an event with a launch or progress update.

3.1 Topic Categories

Class label	Definition	Representative examples
Launches, Platforms & Partnerships	Introduces a digital platform, service, formal collaboration, or technology initiative.	Aarogya Setu 2.0; NHCX; UHI; e-Sushrut Clinic; SAHI/BODH; NHA partnerships.
Events, Hackathons & Conferences	Centres on an event, summit, showcase, hackathon, workshop, conclave, or conference.	Hackathon showcases; digital-health summits; Chintan Shivir; review/conference meetings where the event is the main focus.
Progress Updates & Reviews	Reports a milestone, implementation progress, adoption metric, programme review, or status update.	ABHA-linked health records; Scan & Share registrations; ABHA IDs created; ABDM implementation updates.

3.2 Class Distribution

The 34-document dataset has a more balanced distribution than the original 11-record subset. The Events, Hackathons & Conferences class is largest at 13 records; Launches, Platforms & Partnerships has 11, and Progress Updates & Reviews has 10. The class counts are still small, but all three categories have enough examples for a proportional stratified split.

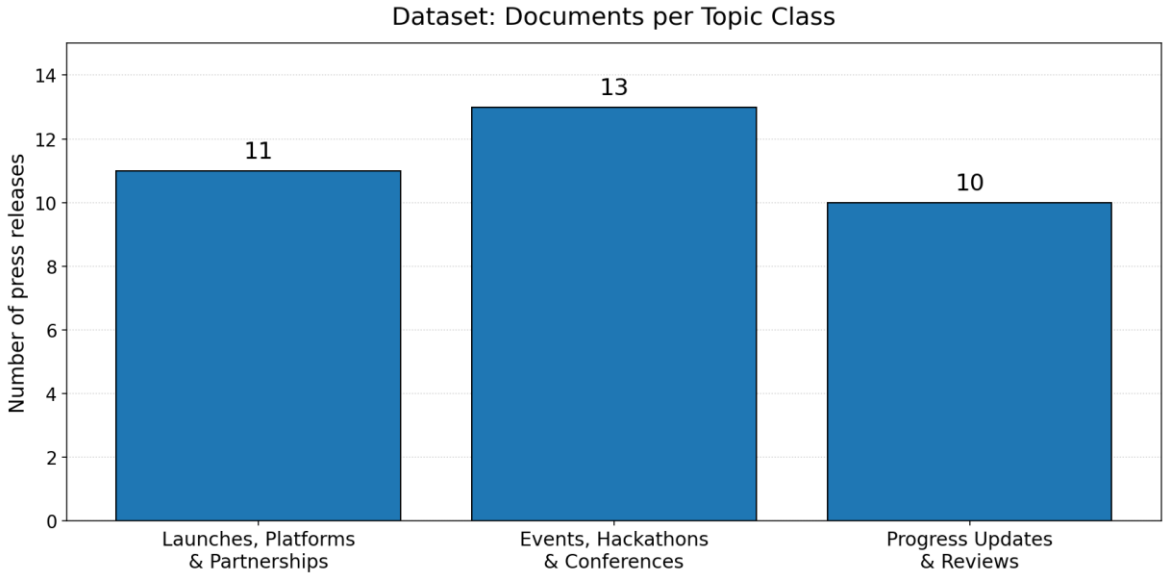


Figure 1. Class counts in the 34-document dataset.

3.3 Data Preparation and Integrity

- The loader reads UTF-8 JSON, tolerates a byte-order mark where present, repairs common escape issues, ignores records that are not dictionaries, and retains entries with usable text and identifiers.
- Whitespace is normalized before training, and the cleaning function removes residual HTML-like markup and URLs while preserving the words that indicate the release topic.
- The dataset stores an explicit label field, reducing dependence on filename-only rules.
- The original JSON file is preserved. Five unmapped records from the earlier input remain separate for manual verification.
- The source-page extraction code uses HTML heuristics; each downloaded article should be checked for truncation, navigation text, repeated headers, and duplicate or near-duplicate content.

3.4 Data Split

The classifier uses a deterministic random seed (42) and proportional stratified splitting. The held-out test partition is created first at 20%; 20% of the remaining records are assigned to validation. For 34 records, the realized sizes are 21 training, 6 validation, and 7 test records. The test set is used only for final evaluation.

Partition	Documents	Purpose
Training	21	Fit TF-IDF + Logistic Regression and update DistilBERT weights.
Validation	6	Monitor DistilBERT during training; it is not the final test set.
Test	7	Final comparison and error analysis after training.
Total	34	Labelled dataset used in the reported run.

A stratified split preserves class representation as far as possible, but seven test examples are still too few for a stable estimate of real-world performance. One correct or incorrect prediction changes accuracy by about 14.3 percentage points.

4. System Architecture

The workflow has two complementary branches after data loading and label preparation. A classical NLP branch provides interpretable linguistic analysis. A modelling branch trains a sparse TF-IDF baseline and a contextual transformer classifier on the same train/validation/test split. Both make one class prediction per press release.

4.1 End-to-End Processing Flow

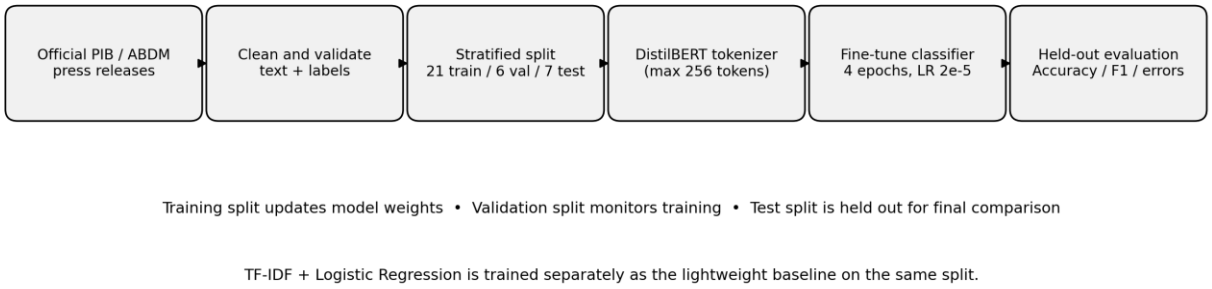


Figure 2. Dataset-to-evaluation flow for the fine-tuned DistilBERT classifier. The TF-IDF baseline is trained separately on the same split.

4.2 Core Modules

Module	Responsibility
Input loader	Reads and validates JSON records, repairs common export issues, and normalizes whitespace.
Dataset builder	Retrieves official PIB text, records metadata and starter labels, and generates the JSON dataset.
Text cleaner	Removes residual markup and URLs and collapses repeated whitespace.
Classical NLP analysis	Runs sentence tokenization, tokenization, POS tagging, POS-aware lemmatization, morphological inspection, and WordNet lookup.
Label mapper	Maintains one of the three topic labels for each usable press release.
Baseline classifier	Fits TF-IDF unigrams/bigrams and Logistic Regression on training text only.
Transformer classifier	Tokenizes releases and fine-tunes DistilBERT with a three-class classification head.
Evaluation	Calculates accuracy, weighted precision/recall/F1, a per-class report, confusion matrix, and test error previews.

4.3 Libraries and APIs

Library / Tool	Role in the project
Python, json, re, pathlib	Data handling, JSON parsing, text normalization, and file management.
requests, BeautifulSoup	Retrieving official web pages and extracting article text in the dataset preparation workflow.
NLTK	Sentence/word tokenization, POS tagging, stop-word lists, Porter stemming, WordNet lemmatization and corpus access.
WordNet	Lexical synsets, definitions, synonyms, and hypernym relations for semantic inspection.
spaCy (en_core_web_sm)	Morphological features, dependency/phrase analysis, and named entity recognition in supplementary experiments.
scikit-learn	TfidfVectorizer, LogisticRegression, train_test_split, accuracy, precision/recall/F1, classification report, confusion matrix.
Hugging Face Transformers	AutoTokenizer, AutoModelForSequenceClassification, Trainer, TrainingArguments, model inference.
Hugging Face Datasets	Conversion of text/label partitions into datasets used by the Trainer.
PyTorch and NumPy	Tensor operations, random seeds, model training backend, and metric arrays.
Matplotlib	Dataset class-count chart, model metric comparisons, and training/evaluation visualizations.

5. NLP Preprocessing Pipeline

Preprocessing makes the source text more consistent and enables inspection of the linguistic properties of the release. The full cleaned document is retained for classification. Linguistic transformations are mainly used for explanatory analysis and are not silently appended as extra features to the main TF-IDF or DistilBERT input.

5.1 Text Cleaning and Noise Handling

The cleaning stage removes HTML-like markup and URLs, replaces repeated whitespace with a single space, and trims leading/trailing whitespace. The practical experiments also demonstrate special-character handling, contraction expansion, lowercasing, and number removal as separate variants. Number removal is not automatically applied to the main classifier text because figures such as ABHA counts, registrations, and dates may be meaningful evidence for progress-oriented releases.

5.2 Sentence Boundary Identification

NLTK sentence tokenization splits the cleaned document into sentence units for inspection and for later tasks such as summarization. On the representative Aarogya Setu 2.0 release used in the earlier preprocessing run, the output reported 33 sentences. Press releases often include title, subtitle, posting timestamp, section headers, and long bullet-like blocks; these may produce imperfect sentence boundaries and should be reviewed when the extraction is noisy.

5.3 Tokenization

Word tokenization divides text into words and punctuation tokens. The representative release produced 1,240 tokens. A short token sample includes “Ministry”, “of”, “Health”, “and”, “Family”, “Welfare”, “Union”, “Health”, “Minister”, “Shri”, “J.P.”, “Nadda”, “to”, “Launch”, and “Aarogya”. Tokenization is a prerequisite for POS tagging, lemmatization, phrase extraction, and lexical semantic analysis.

5.4 Stop-word Handling and Number Preservation

The notebook demonstrates English stop-word removal for lexical analyses such as n-grams and embeddings. For the primary classifier, stop-word removal is applied inside the TF-IDF vectorizer, while the DistilBERT text is cleaned but remains largely intact. This distinction matters: removing common words can help sparse lexical models, but transformer models may use word order, grammatical cues, and function words. Similarly, numeric counts can signal progress and should generally be retained unless a specific experiment tests their removal.

5.5 POS Tagging and Syntactic Analysis

NLTK POS tagging assigns fine-grained Penn Treebank tags such as NNP (proper noun), NN (singular/common noun), VB/VBD (verb), JJ (adjective), IN (preposition), and CD (number). A sample from the launch release tags “Ministry”, “Health”, “Family”, and “Welfare” as proper nouns, and the verb “Launch” as VB. These tags are useful for linguistic inspection, but they can confuse acronyms, brand names, and government programme terms. In the core classification pipeline, POS tags are not concatenated to the model input.

5.6 Lemmatization and Morphological Analysis

The POS-aware lemmatizer maps NLTK tags to WordNet categories (adjective, verb, noun, or adverb) before reducing inflected forms to their base forms. In the representative release, “initiatives” becomes “initiative” and “standards” becomes “standard”. The dedicated morphology experiment uses spaCy to expose token lemmas and features; for example, “is” is reduced to “be” and tagged as an auxiliary verb with present-tense, singular, third-person features.

5.7 WordNet Semantic Analysis

WordNet maps words to synsets—sets of related word senses with definitions and semantic relations. The project retrieves example synsets, definitions, synonyms, and hypernyms. For “health”, one sense is “a healthy state of wellbeing free from disease” with hypernym “wellbeing.n.01”. WordNet is useful for demonstrating lexical semantics, but the first synset may not be the meaning intended in a government announcement. It also has limited coverage of brand names and programme acronyms, so its output is interpretive rather than a direct class feature.

5.8 Phrase Extraction, N-grams, and Representations

The practical pipeline demonstrates bigrams/trigrams as adjacent-word statistics and a POS-based noun-phrase chunker for grammatical phrases. It also demonstrates TF-IDF, Word2Vec, and DistilBERT embeddings. These experiments explain alternative ways to represent text: TF-IDF creates sparse document-term vectors, Word2Vec learns static dense word vectors from context, and DistilBERT creates contextual subword vectors. In the core classifier, TF-IDF is used for the baseline and DistilBERT embeddings are learned through the fine-tuned sequence-classification model; the Word2Vec demo is not a feature of the main classifier.

5.9 Related Laboratory Experiments

Experiment	Relevance to this project
3. Basic pre-processing	HTML/special-character removal, whitespace cleaning, contractions, lowercasing, numbers, sentence segmentation, tokens, stop words, n-grams, noun phrases, ASCII validation.
4. Advanced pre-processing	Stemming, POS-aware lemmatization, noise handling, multilingual-processing considerations, entity anonymization.
5. Shallow and deep parsing	POS-based chunking, CFG parse trees, and dependency relations.
6. Text processing models	N-gram counts, TF-IDF, Word2Vec and DistilBERT contextual embeddings.
7. Morphological analysis	Lemmas, POS tags, and grammatical features such as number, tense and person.
8. HMM POS tagging	A separate sequence-tagging demonstration using an HMM/Viterbi model; not the primary document classifier.
9. NER and WordNet	Named entity identification and semantic lookup; helps illustrate domain vocabulary limitations.
10. Zero-shot classification	Supplementary pretrained MNLI classification demonstration; distinct from the supervised three-class evaluation reported in Sections 6–8.

6. Classification Models

6.1 Baseline: TF-IDF + Logistic Regression

The baseline converts each press release into a sparse feature vector using TfidfVectorizer. The classifier allows up to 5,000 features, uses unigrams and bigrams, and removes English stop words in this vectorizer. Logistic Regression is trained with a maximum of 1,000 iterations, balanced class weights, and the fixed random seed. This approach is computationally lightweight and provides an interpretable benchmark: feature weights can be inspected to understand which words or short phrases influence a prediction.

TF-IDF weights terms by combining how frequently they occur in one document with how widely they occur across the collection. Unigrams capture individual terms such as “launch” or “hackathon”; bigrams capture short phrases such as “health records” or “digital platform”. Because the vectorizer is fitted on the training partition only, the held-out test documents do not influence the feature vocabulary.

6.2 Transformer: DistilBERT

The second model uses the pretrained distilbert-base-uncased checkpoint through Hugging Face Transformers. A sequence-classification head is configured for the three project labels. The tokenizer pads and truncates documents to a maximum of 256 tokens. The model is fine-tuned on the training set, with the validation set used for epoch-wise evaluation. Fine-tuning updates the pretrained model parameters for the project categories; it is not the same as training a large language model from random initialization.

A 256-token input limit makes training more manageable but can omit content near the end of long press releases. This matters because the key milestone or conclusion may occur after the title and first few paragraphs. More data, chunking, or longer input windows are potential future improvements, but should be tested rather than assumed to improve results.

6.3 Training Configuration

Parameter	Configured value	Purpose
Dataset	final_expanded.json (34 records)	Explicit text and starter label fields.
Pretrained checkpoint	distilbert-base-uncased	Base encoder for sequence classification.
Number of classes	3	One output class per release topic.
Random seed	42	Repeatability of partitions and training initialization.
Epochs	4	Training passes over the training subset.
Learning rate	2 x 10^-5	Fine-tuning update size.
Train/eval batch size	2 / 2	Examples processed per batch.
Maximum sequence length	256 tokens	Long documents are truncated.
Weight decay	0.01	Regularization during fine-tuning.
Split	21 train / 6 validation / 7 test	Proportional stratified split.
Baseline features	TF-IDF, unigrams + bigrams, max 5,000	Sparse lexical representation.

6.4 Fine-tuning Process

The fine-tuning sequence required by the project guideline is shown in Figure 2. The training split updates model weights; validation data monitors behaviour during training; and the test split is held aside for the final model comparison. This separation reduces the risk of reporting a score that has been optimized directly on the test documents.

6.5 Supplementary Zero-Shot Experiment

A separate Experiment 10 uses the pretrained typeform/distilbert-base-uncased-mnli pipeline to classify a manually provided sentence against candidate labels without task-specific training examples. For the sentence "Ayushman App and e-Sushrut Clinic strengthen India's Digital Public Infrastructure for Health.", the recorded highest score was "healthcare technology" at 0.4993. This is a separate illustrative zero-shot demonstration with a different candidate-label set; its score is not mixed into the supervised three-class metrics in this report.

7. Evaluation Methodology

Evaluation compares predicted labels with the labelled test examples. Because each release has exactly one reference category, the task is single-label multiclass classification. The primary quantitative results are accuracy, weighted precision, weighted recall, and weighted F1. The code also prints a per-class classification report, a confusion matrix, and an error preview for each held-out test document.

Metric	Interpretation
Accuracy	Proportion of test releases assigned the correct label.
Weighted precision	Per-class precision averaged by the number of true examples in each class.
Weighted recall	Per-class recall averaged by class support.
Weighted F1	Weighted average of per-class F1; balances precision and recall.
Per-class report	Shows precision, recall, F1 and support for each of the three labels.
Confusion matrix	Counts actual-vs-predicted labels and helps reveal which classes are confused.
Error analysis	Prints actual and predicted labels with a text preview to support qualitative review.

7.1 Why BERTScore Is Not the Primary Metric

The course guideline lists BERTScore and F1-score as examples of evaluation metrics. For this project, F1-score and the other classification metrics are the appropriate primary measures because the system predicts one discrete category label rather than generating a sentence or summary. BERTScore is designed to compare semantic similarity between generated text and reference text, so it is not needed for the main classification evaluation. It could be relevant if a future extension generated summaries or free-form explanations.

7.2 Test-set Discipline and Reproducibility

- Fit the TF-IDF vectorizer on training text only; transform validation/test text without refitting.
- Use the validation partition for training-time monitoring, not for final headline performance.
- Evaluate against the test partition once model settings are chosen; avoid repeatedly tuning to the same seven items.
- Record the dataset version, model checkpoint, random seed, parameters, full console output, and each test prediction.
- After adding more records to the dataset, repeat evaluation across multiple seeds or use cross-validation to better estimate variability.

8. Results and Discussion

8.1 Dataset Output

The 34-record run reported 34 labelled releases. The dataset distribution was 11 Launches, Platforms & Partnerships; 13 Events, Hackathons & Conferences; and 10 Progress Updates & Reviews. The stratified partition contained 21 training documents, 6 validation documents, and 7 test documents. Although the current dataset contains more records than the initial 11-document baseline, the collection remains small for fine-tuning and robust final evaluation.

8.2 Model Results on the Dataset Split

Model	Accuracy	Weighted precision	Weighted recall	Weighted F1
TF-IDF + Logistic Regression	0.7143 (71.43%)	0.8286 (82.86%)	0.7143 (71.43%)	0.7024 (70.24%)
Fine-tuned DistilBERT	0.4286 (42.86%)	0.3095 (30.95%)	0.4286 (42.86%)	0.3592 (35.92%)



Figure 3. Held-out metrics reported for the 34-document run (test n = 7).

On this split, the TF-IDF baseline made the stronger set of predictions, with five of seven test documents correct (71.43% accuracy). DistilBERT classified three of seven correctly (42.86%). The weighted F1 difference was approximately 34.32 percentage points in favour of the baseline. This does not demonstrate that TF-IDF is generally superior; it shows that, on this small dataset and chosen split, simple lexical features separated the examples more reliably than the fine-tuned transformer.

8.3 Comparison with the Original 11-Document Run

Run	Corpus / test size	Model	Accuracy	Weighted F1
Original baseline	11 documents / 3 test	TF-IDF + Logistic Regression	33.33%	16.67%
Current TF-IDF run	34 documents / 7 test	TF-IDF + Logistic Regression	71.43%	70.24%
Current DistilBERT run	34 documents / 7 test	Fine-tuned DistilBERT	42.86%	35.92%

The comparison between the original and current baseline is encouraging, but it is not a controlled same-test-set comparison: the number of training examples, test examples, and sampled releases changed. The result should be described as the outcome of a newer dataset run, not as a causal estimate of the effect of adding records alone.

8.4 Class-level Errors and Interpretation

The current run's DistilBERT per-class report showed zero precision, recall, and F1 for Progress Updates & Reviews. In practical terms, it did not correctly predict a test item as belonging to that class. The error previews included examples such as a release about more than 52.5 crore ABHA IDs being assigned to Launches, a PM-JAY/ABDM National Review Meeting being assigned to Events, and a SAHI/BODH launch being assigned to Events. These mistakes highlight the overlap between surface words and the main purpose of a release.

A launch announcement can also report adoption progress; a meeting can be an event while its central purpose is to review implementation; and a document can mention a new technology without its primary label being "launch". Annotation rules should prioritize the main news purpose rather than a single trigger word. The data should include several examples of each ambiguous pattern before model tuning is attempted.

8.5 Interpretation of Scores

The weighted metrics summarize performance across classes while respecting class support, but they do not remove uncertainty from having only seven test documents. An accuracy change of one test document changes the score by about 14.3 percentage points. Weighted precision can be higher or lower than accuracy depending on the prediction distribution. Transformer softmax scores are confidence-like model outputs and should not be treated as calibrated probabilities unless a calibration study is performed.

9. Application and Live Demonstration

The system supports two usage modes. Batch mode loads and labels the stored corpus, runs preprocessing, trains both classifiers, and reports the held-out metrics. Live mode accepts a new paragraph and returns a predicted category. The live demo helps demonstrate the application to an audience, but it must be described separately from the held-out evaluation results.

9.1 Sample Live Input

> The Ministry of Health and Family Welfare launched a new digital health platform to enable citizens to access healthcare services online. The platform brings together hospitals, health records and digital health services, making healthcare more accessible and convenient for citizens across the country.

9.2 Output

Predicted class: Launches, Platforms & Partnerships.

This is a sensible category because the central action is the launch of a digital platform. Live input text is not included in the test metrics unless it is labelled in advance and placed in a valid evaluation protocol. It should not be presented as an additional correct test prediction.

9.3 Role of Supplementary NLP Experiments

Experiments 3–9 demonstrate the course-required linguistic processes and supporting models. Experiment 10 demonstrates zero-shot classification with a pretrained MNLI pipeline. Together, these experiments provide practical evidence that the team has explored preprocessing, syntactic and semantic analysis, embeddings, NER, and transformer inference. The core MPR result, however, remains the supervised three-class classification experiment evaluated on the 34-document dataset.

10. Limitations, Risks, and Future Scope

10.1 Current Limitations

- Small corpus: 34 documents are still too few to estimate generalization across all government health-policy releases; the test set contains only seven examples.
- Starter labels need expert review: categories were assigned from each release's main subject and ambiguous launch/event/progress releases can be interpreted differently.
- Transformer underperformance in the current run: DistilBERT produced weaker weighted F1 than the TF-IDF baseline and failed to identify Progress Updates & Reviews in the test set.
- Long-document truncation: the DistilBERT classifier sees at most 256 tokens, so content later in the release may be omitted.
- Domain terminology: general-purpose POS taggers, NER models, and WordNet may misinterpret ministry acronyms, product names, health programmes, and specialized concepts.
- Source extraction quality: HTML extraction can include navigation text, malformed sections, or duplicated boilerplate unless each record is inspected.
- Single split variability: the reported metrics depend on the particular train/validation/test partition; repeated splits are needed for more stable estimates.

10.2 Future Improvements

- Increase the dataset to at least dozens of reviewed releases per class, including multiple years, schemes, event types, and writing styles.
- Use written annotation rules and have a second reviewer label ambiguous records; report agreement and document adjudication decisions.
- Inspect every scraped release for completeness, remove exact/near duplicates, and keep an auditable source URL and date.
- Use repeated stratified splits or cross-validation after enough labelled data is available.
- Compare TF-IDF + Logistic Regression with linear SVM, Naive Bayes, and other simple baselines before adding model complexity.
- Test longer token limits or chunking with an aggregation rule if early truncation removes useful information.
- Consider class weighting, balanced sampling, early stopping, and careful hyperparameter search for DistilBERT, using only training/validation data.
- Add a small human-reviewed error set focused on mixed-purpose releases, and revise label definitions where annotators disagree.

- Expose an inference interface that accepts text, returns the predicted class and confidence-like score, and links back to the original release.

10.3 Responsible Use

The classifier should be treated as an assistive organization tool rather than an authoritative interpretation of government policy. A predicted category does not establish the truth, importance, quality, or policy implications of the announcement. Users should inspect the original release when classification informs research, reporting, or public communication. The source text and metadata should remain traceable, and sensitive personal information should not be added to training data without an appropriate basis.

11. Conclusion

This mini project presents an end-to-end NLP workflow for categorizing official government health-policy press releases into three topic classes. It combines data validation and source collection with classical linguistic analysis, a TF-IDF + Logistic Regression baseline, and fine-tuning of DistilBERT for sequence classification. The report documents the required preprocessing operations—sentence boundary identification, tokenization, lemmatization, POS tagging, and WordNet semantic lookup—while clarifying which components directly feed the classifier and which are supplementary demonstrations.

The current run contained 34 labelled releases, distributed as 11 Launches, Platforms & Partnerships, 13 Events, Hackathons & Conferences, and 10 Progress Updates & Reviews. On a seven-document held-out test set, TF-IDF + Logistic Regression achieved 71.43% accuracy and 70.24% weighted F1, while fine-tuned DistilBERT achieved 42.86% accuracy and 35.92% weighted F1. These metrics are preliminary because the data set is small, the labels need further review, and one example changes the accuracy noticeably. The strongest current finding is therefore methodological: a larger and carefully audited dataset plus transparent baseline comparison provides a more informative starting point than relying on a transformer model alone. Larger reviewed data and repeated evaluation are required before claiming reliable real-world performance.

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Appendix A. Practical Labeling Rules

Use the main purpose of the release, not a single keyword, to assign the category. The following checklist is intended to make manual annotation consistent:

- Launches, Platforms & Partnerships: choose this when the main news is that a service/platform/technology is launched, introduced, enabled, or formalized through a partnership/MoU.
- Events, Hackathons & Conferences: choose this when the main news is an event itself—such as a workshop, summit, hackathon, showcase, conclave, or conference. A review meeting may fall here if the event is the main subject; use Progress Updates & Reviews when reported implementation outcomes are the main point.
- Progress Updates & Reviews: choose this when the release primarily reports milestones, adoption counts, implementation status, programme review results, or an update on an existing programme.
- For mixed-purpose releases, read the title and opening paragraphs, then consider the dominant result or action. Record uncertain cases for a second reviewer instead of forcing an undocumented label.

Appendix B. Key Observed Results

Measure	TF-IDF + Logistic Regression	Fine-tuned DistilBERT
Accuracy	71.43%	42.86%
Weighted precision	82.86%	30.95%
Weighted recall	71.43%	42.86%
Weighted F1	70.24%	35.92%
Test items correctly classified	5 of 7	3 of 7
Notable error finding	Higher aggregate metrics on this split.	Progress Updates & Reviews: precision, recall and F1 were 0.00 in this run.